

信息学中的概率统计

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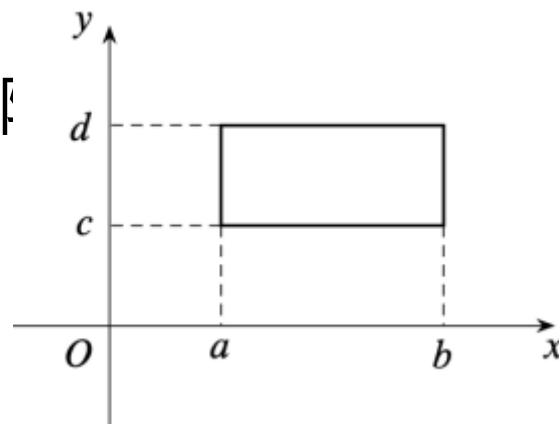
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多维连续随机变量

1. 多维连续随机变量的分布函数和密度函数
2. 多维连续随机变量的独立性
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1. 多维连续随机变量的分布函数和密度函数

- ▶ 给定二维随机变量 X, Y 和实数 x, y , 定义 $F(x, y) = P(X \leq x, Y \leq y)$ 为二维随机变量 X, Y 的**联合分布函数**
- ▶ 性质1(**有界性**): $0 \leq F(x, y) \leq 1, F(-\infty, y) = 0, F(x, -\infty) = 0, F(+\infty, +\infty) = 1$
- ▶ 性质2(**单调性**): $x_1 < x_2 \Rightarrow F(x_1, y) \leq F(x_2, y), y_1 < y_2 \Rightarrow F(x, y_1) \leq F(x, y_2)$
- ▶ 性质3(**右连续**): $F(x+0, y) = F(x, y), F(x, y+0) = F(x, y)$
- ▶ 性质4(**非负性**): $P(a < X \leq b, c < Y \leq d) = F(b, d) - F(a, d) - F(b, c) + F(a, c) \geq 0$
- ▶ $F(x, y)$ 满足上述四条性质**等价于** $F(x, y)$ 是某个二维
- ▶ 性质4是否被性质1-3蕴含?
 - ▶ $F(x, y) = 1$ 当 $x + y \geq 0$, 否则 $F(x, y) = 0$



1. 多维连续随机变量的分布函数和密度函数

- ▶ 给定二维随机变量 X, Y , 若存在 $f(x, y) \geq 0$ 使得分布函数 $F(x, y)$ 可表示为 $F(x, y) = \int_{-\infty}^x \int_{-\infty}^y f(u, v) du dv$, 则称二维随机变量 X, Y 为**二维连续随机变量**, 称 $f(x, y)$ 为 X, Y 的**联合密度函数**
- ▶ 联合密度函数的性质
- ▶ 性质1(**非负性**): $f(x, y) \geq 0$
- ▶ 性质2(**正则性**): $\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(u, v) du dv = 1$
- ▶ 在 $F(x, y)$ 偏导数存在的点, $\frac{\partial^2 F(x, y)}{\partial x \partial y} = f(x, y)$
- ▶ 对于区域 G , $P((X, Y) \in G) = \iint_G f(x, y) dx dy$

1. 多维连续随机变量的分布函数和密度函数

▶ 例1: X, Y 的联合密度函数满足

▶ $f(x, y) = c \cdot e^{-2x-3y}$ 若 $x > 0, y > 0$

▶ 否则 $f(x, y) = 0$

▶ 这里 c 是某个常数

▶ 计算常数 c , 并计算 $P(X < 1, Y > 1), P(X > Y)$

▶ 由正则性 $\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x, y) dx dy = c \cdot \frac{1}{2} \cdot \frac{1}{3} = 1 \Rightarrow c = 6$

▶ $P(X < 1, Y > 1) = \int_0^1 \int_1^{+\infty} 6 \cdot e^{-2x-3y} dy dx = (1 - e^{-2})e^{-3}$

▶ $P(X > Y) = \int_0^{+\infty} \int_0^x 6 \cdot e^{-2x-3y} dy dx = \int_0^{+\infty} 6 \cdot e^{-2x} \cdot \frac{1}{3} \cdot (1 - e^{-3x}) dx = \frac{4}{5}$

1. 多维连续随机变量的分布函数和密度函数

- ▶ 例2：给定 $\mathbb{R}^2$ 中的一个有界区域 D 。随机变量 (X, Y) 表示从 D 中均匀取一点的坐标。写出 X, Y 的联合密度函数。
 - ▶ $f(x, y) = 1/S_D$ 若 $(x, y) \in D$
 - ▶ 否则 $f(x, y) = 0$
- ▶ 称 (X, Y) 服从 D 上的**二维均匀分布**，记为 $(X, Y) \sim U(D)$

1. 多维连续随机变量的分布函数和密度函数

- ▶ 给定二维随机变量 X, Y 的**联合分布函数** $F(x, y)$, 如何计算 X 的分布函数?

- ▶ $P(X \leq x) = P(X \leq x, Y < +\infty) = F(x, +\infty)$

- ▶ 定义 $F_X(x) = F(x, +\infty) = \lim_{y \rightarrow +\infty} F(x, y)$ 为 X 的**边际分布函数**

- ▶ 类似有 $F_Y(y) = F(+\infty, y) = \lim_{x \rightarrow +\infty} F(x, y)$ 为 Y 的**边际分布函数**

- ▶ 例: X, Y 的联合分布函数满足

- ▶ $F(x, y) = 1 - e^{-x} - e^{-y} + e^{-x-y-\lambda xy}$ 当 $x > 0, y > 0$,

- ▶ 否则 $F(x, y) = 0$

- ▶ 参数 $0 \leq \lambda \leq 1$

- ▶ 计算 $F_X(x)$ 和 $F_Y(y)$

- ▶ $F_X(x) = 1 - e^{-x}$ 当 $x > 0$

- ▶ $F_Y(y) = 1 - e^{-y}$ 当 $y > 0$

1. 多维连续随机变量的分布函数和密度函数

▶ 给定二维连续随机变量 X, Y , $f(x, y)$ 为 X, Y 的联合密度函数

▶ 如何计算 X 的密度函数 $f_X(x)$?

▶ $P(X \leq x) = F(x, +\infty) = \int_{-\infty}^x \int_{-\infty}^{+\infty} f(u, v) dv du = \int_{-\infty}^x f_X(u) du$

▶ $f_X(x) = \int_{-\infty}^{+\infty} f(x, y) dy$ 为 X 的**边际密度函数**

▶ 类似 $f_Y(y) = \int_{-\infty}^{+\infty} f(x, y) dx$ 为 Y 的**边际密度函数**

▶ 例1: X, Y 的联合密度函数满足

▶ $f(x, y) = 6 \cdot e^{-2x-3y}$ 若 $x > 0, y > 0$

▶ 否则 $f(x, y) = 0$

▶ 计算 X, Y 的边际密度函数

▶ $f_X(x) = \int_{-\infty}^{+\infty} f(x, y) dy = 2 \cdot e^{-2x}$

▶ $f_Y(y) = \int_{-\infty}^{+\infty} f(x, y) dx = 3 \cdot e^{-3y}$

1. 多维连续随机变量的分布函数和密度函数

- ▶ $f_X(x) = \int_{-\infty}^{+\infty} f(x, y)dy$ 为 X 的边际密度函数
- ▶ $f_Y(y) = \int_{-\infty}^{+\infty} f(x, y)dx$ 为 Y 的边际密度函数

- ▶ 例2: X, Y 的联合密度函数满足
 - ▶ $f(x, y) = 1$ 若 $0 < x < 1, |y| < x$
 - ▶ 否则 $f(x, y) = 0$
- ▶ 计算 Y 的边际密度函数
 - ▶ 当 $|y| > 1$, $f_Y(y) = 0$
 - ▶ 当 $-1 < y < 0$, $-y < x < 1$, $f_Y(y) = \int_{-\infty}^{+\infty} f(x, y)dx = \int_{-y}^1 f(x, y)dx = 1 + y$
 - ▶ 当 $0 < y < 1$, $y < x < 1$, $f_Y(y) = \int_{-\infty}^{+\infty} f(x, y)dx = \int_y^1 f(x, y)dx = 1 - y$

1. 多维连续随机变量的分布函数和密度函数

- ▶ 回顾：若 $P(Y = y_j) > 0$ ，则称 $p_{i|j} = P(X = x_i | Y = y_j) = P(X = x_i, Y = y_j) / P(Y = y_j)$ 为给定 $Y = y_j$ 条件下 X 的**条件分布列**
- ▶ 对于连续随机变量，如何定义条件分布函数和条件密度函数？
- ▶ $P(X \leq x | Y = y)$ 可定义为 $\lim_{\Delta \rightarrow 0} P(X \leq x | y \leq Y \leq y + \Delta)$
- ▶ $P(X \leq x | Y = y) = \lim_{\Delta \rightarrow 0} P(X \leq x | y \leq Y \leq y + \Delta) = \lim_{\Delta \rightarrow 0} \frac{P(X \leq x, y \leq Y \leq y + \Delta)}{P(y \leq Y \leq y + \Delta)}$
- ▶ $\lim_{\Delta \rightarrow 0} \frac{P(X \leq x, y \leq Y \leq y + \Delta)}{P(y \leq Y \leq y + \Delta)} = \lim_{\Delta \rightarrow 0} \frac{\int_{-\infty}^x \left(\frac{1}{\Delta} \int_y^{y+\Delta} f(u, v) dv \right) du}{\frac{1}{\Delta} \int_y^{y+\Delta} f_Y(v) dv}$

1. 多维连续随机变量的分布函数和密度函数

► 当密度函数连续

$$\text{► } \lim_{\Delta \rightarrow 0} \frac{1}{\Delta} \int_y^{y+\Delta} f(u, v) dv = f(u, y)$$

$$\text{► } \lim_{\Delta \rightarrow 0} \frac{1}{\Delta} \int_y^{y+\Delta} f_Y(v) dv = f_Y(y)$$

$$\text{► } P(X \leq x | Y = y) = \lim_{\Delta \rightarrow 0} \frac{\int_{-\infty}^x \left(\frac{1}{\Delta} \int_y^{y+\Delta} f(u, v) dv \right) du}{\frac{1}{\Delta} \int_y^{y+\Delta} f_Y(v) dv} = \frac{\int_{-\infty}^x f(u, y) du}{f_Y(y)} = \int_{-\infty}^x \frac{f(u, y)}{f_Y(y)} du$$

$$\text{► } F(x|y) = P(X \leq x | Y = y) = \int_{-\infty}^x \frac{f(u, y)}{f_Y(y)} du \text{ 为给定 } Y = y \text{ 条件下 } X \text{ 的} \textbf{条件分布函数}$$

$$\text{► } f(x|y) = \frac{f(x, y)}{f_Y(y)} \text{ 为给定 } Y = y \text{ 条件下 } X \text{ 的} \textbf{条件密度函数}$$

► 正则性?

$$\text{► } \int_{-\infty}^{+\infty} \frac{f(x, y)}{f_Y(y)} dx = \frac{\int_{-\infty}^{+\infty} f(x, y) dx}{f_Y(y)} = \frac{f_Y(y)}{f_Y(y)} = 1$$

1. 多维连续随机变量的分布函数和密度函数

- ▶ 例：随机变量 X, Y 服从单位圆 ($x^2 + y^2 \leq 1$)上的二维均匀分布。计算 $f(x|y)$
- ▶ $f(x|y) = \frac{f(x,y)}{f_Y(y)}$
- ▶ $f(x, y) = \frac{1}{\pi}$ 若 (x, y) 在单位圆内
- ▶ 否则 $f(x, y) = 0$
- ▶ $f_Y(y) = \int_{-\infty}^{+\infty} f(x, y)dx = \int_{-\sqrt{1-y^2}}^{+\sqrt{1-y^2}} f(x, y)dx = \frac{2\sqrt{1-y^2}}{\pi}$
- ▶ $f(x|y) = \frac{1}{\pi} / \frac{2\sqrt{1-y^2}}{\pi} = \frac{1}{2\sqrt{1-y^2}}$ 若 $|x| \leq \sqrt{1-y^2}$
- ▶ 否则 $f(x|y) = 0$

2. 多维连续随机变量的独立性

- ▶ 回顾：给定二维离散随机变量 (X, Y) ，若对于任意实数 x, y 均有 $P(X = x, Y = y) = P(X = x) \cdot P(Y = y)$ ，则称 X, Y **相互独立**
- ▶ 给定二维随机变量 (X, Y) ，分布函数为 $F(x, y)$ ，边缘分布函数为 $F_X(x)$ 和 $F_Y(y)$ 。若对于任意实数 x, y 均有 $F(x, y) = F_X(x) \cdot F_Y(y)$ ，则称 X, Y **相互独立**
- ▶ 对于离散随机变量，相互独立**等价于**对于任意实数 x, y 均有 $P(X = x, Y = y) = P(X = x) \cdot P(Y = y)$
- ▶ 对于连续随机变量，相互独立**等价于**对于任意实数 x, y 均有密度函数 $f(x, y) = f_X(x) \cdot f_Y(y)$

2. 多维连续随机变量的独立性

- ▶ 例1: X, Y 的联合密度函数满足
 - ▶ $f(x, y) = 6 \cdot e^{-2x-3y}$ 若 $x > 0, y > 0$
 - ▶ 否则 $f(x, y) = 0$
- ▶ 判断 X, Y 是否相互独立
 - ▶ $f_X(x) = 2 \cdot e^{-2x}$
 - ▶ $f_Y(y) = 3 \cdot e^{-3y}$
 - ▶ $f(x, y) = f_X(x) \cdot f_Y(y)$

2. 多维连续随机变量的独立性

- ▶ 例2：令随机变量 X 某服务器第一次发生故障的时间， Y 表示另一台服务器第一次发生故障的时间。已知则 $X \sim \text{Exp}(\lambda_1)$ ， $Y \sim \text{Exp}(\lambda_2)$ ，且 X 与 Y 相互独立。
- ▶ 计算 $P(X < Y)$

- ▶
$$P(X < Y) = \int_{x=0}^{+\infty} \int_{y=x}^{+\infty} f(x, y) dx dy = \int_{x=0}^{+\infty} \int_{y=x}^{+\infty} f_X(x) \cdot f_Y(y) dx dy$$

- ▶
$$f_X(x) \cdot f_Y(y) = \lambda_1 e^{-\lambda_1 x} \cdot \lambda_2 e^{-\lambda_2 y}$$

- ▶
$$P(X < Y) = \int_0^{+\infty} \lambda_1 e^{-\lambda_1 x} \cdot \int_x^{+\infty} \lambda_2 e^{-\lambda_2 y} dy dx = \int_0^{+\infty} \lambda_1 e^{-\lambda_1 x} \cdot P(Y \geq x) dx$$

- ▶
$$P(Y \geq x) = e^{-\lambda_2 x}$$

- ▶
$$P(X < Y) = \int_{x=0}^{+\infty} \lambda_1 e^{-\lambda_1 x} \cdot e^{-\lambda_2 x} dx = \frac{\lambda_1}{\lambda_1 + \lambda_2}$$

2. 多维连续随机变量的独立性

- ▶ 二维连续随机变量 X, Y 的联合密度函数满足

- ▶
$$f(x, y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp \left[-\frac{1}{2(1-\rho^2)} \cdot \left(\frac{(x-\mu_1)^2}{\sigma_1^2} + \frac{(y-\mu_2)^2}{\sigma_2^2} - \frac{2\rho(x-\mu_1)(y-\mu_2)}{\sigma_1\sigma_2} \right) \right]$$

- ▶ 其中 $\mu_1, \mu_2 \in \mathbb{R}, \sigma_1, \sigma_2 > 0, |\rho| < 1$

- ▶ 称 X, Y 服从参数为 $\mu_1, \mu_2, \sigma_1, \sigma_2, \rho$ 的 **二维正态（高斯）分布**

- ▶ 记号： $X, Y \sim N(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$

- ▶ 验证正则性： $\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x, y) dx dy = 1$

- ▶ 计算边际密度函数

- ▶ 计算条件密度函数

- ▶ 判断 X, Y 是否相互独立

2. 多维连续随机变量的独立性

- ▶ $f(x, y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp \left[-\frac{1}{2(1-\rho^2)} \cdot \left(\frac{(x-\mu_1)^2}{\sigma_1^2} + \frac{(y-\mu_2)^2}{\sigma_2^2} - \frac{2\rho(x-\mu_1)(y-\mu_2)}{\sigma_1\sigma_2} \right) \right]$
- ▶ 计算 $\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x, y) dx dy$?
- ▶ 换元法：定义 u', v' ，使得 $\frac{(x-\mu_1)^2}{\sigma_1^2} + \frac{(y-\mu_2)^2}{\sigma_2^2} - \frac{2\rho(x-\mu_1)(y-\mu_2)}{\sigma_1\sigma_2} = (u')^2 + (v')^2$
 - ▶ 配方得： $u' = \frac{x-\mu_1}{\sigma_1} - \rho \cdot \frac{y-\mu_2}{\sigma_2} \Rightarrow v' = \frac{y-\mu_2}{\sigma_2} \cdot \sqrt{1-\rho^2}$
- ▶ 若 $u = \frac{\left(\frac{x-\mu_1}{\sigma_1} - \rho \cdot \frac{y-\mu_2}{\sigma_2} \right)}{\sqrt{1-\rho^2}}, v = \frac{y-\mu_2}{\sigma_2}$ ，则有 $f(x, y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp \left[-\frac{1}{2} (u^2 + v^2) \right]$
- ▶ $\left| \frac{\partial(u, v)}{\partial(x, y)} \right| = \begin{vmatrix} \frac{1}{\sigma_1\sqrt{1-\rho^2}} & -\frac{\rho}{\sigma_2\sqrt{1-\rho^2}} \\ 0 & \frac{1}{\sigma_2} \end{vmatrix} = \frac{1}{\sigma_1\sigma_2\sqrt{1-\rho^2}} \Rightarrow \left| \frac{\partial(x, y)}{\partial(u, v)} \right| = \sigma_1\sigma_2\sqrt{1-\rho^2}$

2. 多维连续随机变量的独立性

- ▶ $f(x, y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left[-\frac{1}{2}(u^2 + v^2)\right]$
- ▶ $\left|\frac{\partial(x,y)}{\partial(u,v)}\right| = \sigma_1\sigma_2\sqrt{1-\rho^2}$
- ▶ $\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x, y) dx dy = \iint \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left[-\frac{1}{2}(u^2 + v^2)\right] \cdot \sigma_1\sigma_2\sqrt{1-\rho^2} du dv$
- ▶ $= \iint \frac{1}{2\pi} \exp\left[-\frac{1}{2}(u^2 + v^2)\right] du dv$
- ▶ $= \int_{-\infty}^{-\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} du \cdot \int_{-\infty}^{-\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{v^2}{2}} dv = 1$
- ▶ 思考：如何从随机变量的角度理解换元 $u = \frac{\left(\frac{x-\mu_1}{\sigma_1} - \rho \cdot \frac{y-\mu_2}{\sigma_2}\right)}{\sqrt{1-\rho^2}}$, $v = \frac{y-\mu_2}{\sigma_2}$?

2. 多维连续随机变量的独立性

- ▶ 边际密度函数 $f_X(x) = \int_{-\infty}^{+\infty} f(x, y) dy$
- ▶
$$f(x, y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp \left[-\frac{1}{2(1-\rho^2)} \cdot \left(\frac{(x-\mu_1)^2}{\sigma_1^2} + \frac{(y-\mu_2)^2}{\sigma_2^2} - \frac{2\rho(x-\mu_1)(y-\mu_2)}{\sigma_1\sigma_2} \right) \right]$$
- ▶ 换元 $u = \frac{\left(\frac{y-\mu_2}{\sigma_2} - \rho \cdot \frac{x-\mu_1}{\sigma_1} \right)}{\sqrt{1-\rho^2}}, v = \frac{x-\mu_1}{\sigma_1}$
- ▶
$$f(x, y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp \left(-\frac{u^2}{2} \right) \cdot \exp \left(-\frac{v^2}{2} \right)$$
- ▶
$$f_X(x) = \int_{-\infty}^{+\infty} f(x, y) dy = \sqrt{1-\rho^2} \cdot \sigma_2 \cdot \int_{-\infty}^{+\infty} f(x, y) du$$
- ▶
$$f_X(x) = \frac{1}{2\pi\sigma_1} \int_{-\infty}^{+\infty} \exp \left(-\frac{u^2}{2} \right) du \cdot \exp \left(-\frac{v^2}{2} \right) = \frac{1}{\sqrt{2\pi} \cdot \sigma_1} \exp \left(-\frac{(x-\mu_1)^2}{2\sigma_1^2} \right)$$
- ▶ 类似有 $f_Y(y) = \frac{1}{\sqrt{2\pi} \cdot \sigma_2} \exp \left(-\frac{(y-\mu_2)^2}{2\sigma_2^2} \right)$

2. 多维连续随机变量的独立性

- ▶ 若 $X, Y \sim N(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$, 则 $X \sim N(\mu_1, \sigma_1^2)$, $Y \sim N(\mu_2, \sigma_2^2)$
 - ▶ 边际分布与参数 ρ 无关 $\Rightarrow$ 具有相同边际分布的多维联合分布可以不同
- ▶ 计算条件密度函数 $f(x|y) = \frac{f(x,y)}{f_Y(y)}$
 - ▶ $f(x, y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp \left[-\frac{1}{2(1-\rho^2)} \cdot \left(\frac{(x-\mu_1)^2}{\sigma_1^2} + \frac{(y-\mu_2)^2}{\sigma_2^2} - \frac{2\rho(x-\mu_1)(y-\mu_2)}{\sigma_1\sigma_2} \right) \right]$
 - ▶ $f_Y(y) = \frac{1}{\sqrt{2\pi}\cdot\sigma_2} \exp \left(-\frac{(y-\mu_2)^2}{2\sigma_2^2} \right)$
 - ▶ $f(x|y) = \frac{1}{\sqrt{2\pi}\sigma_1\sqrt{1-\rho^2}} \exp \left(-\frac{1}{2(1-\rho^2)} \cdot \left(\frac{x-\mu_1}{\sigma_1} - \rho \cdot \frac{y-\mu_2}{\sigma_2} \right)^2 \right)$
 - ▶ $= \frac{1}{\sqrt{2\pi}\sigma_1\sqrt{1-\rho^2}} \exp \left(-\frac{1}{2(1-\rho^2)} \cdot \left(\frac{x-\mu_1 - \rho \cdot \frac{\sigma_1}{\sigma_2} \cdot (y-\mu_2)}{\sigma_1} \right)^2 \right)$
- ▶ 给定 $Y = y$ 条件下, X 的条件分布服从 $N(\mu_1 + \rho \cdot \frac{\sigma_1}{\sigma_2} \cdot (y - \mu_2), \sigma_1^2 (1 - \rho^2))$

2. 多维连续随机变量的独立性

- ▶ 给定 $Y = y$ 条件下, X 的条件分布服从 $N(\mu_1 + \rho \cdot \frac{\sigma_1}{\sigma_2} \cdot (y - \mu_2), \sigma_1^2 (1 - \rho^2))$
- ▶ 给定 $X = x$ 条件下, Y 的条件分布服从 $N(\mu_2 + \rho \cdot \frac{\sigma_2}{\sigma_1} \cdot (x - \mu_1), \sigma_2^2 (1 - \rho^2))$
- ▶ 何时 X, Y 相互独立?
 - ▶ $f(x|y) = \frac{f(x,y)}{f_Y(y)}$
 - ▶ 若 X, Y 相互独立, $f(x, y) = f_X(x) \cdot f_Y(y)$, 也即 $f(x|y) = f_X(x)$
 - ▶ $X \sim N(\mu_1, \sigma_1^2), Y \sim N(\mu_2, \sigma_2^2)$
 - ▶ X, Y 相互独立等价于 $\rho = 0$

3. 多维连续随机变量的特征数

- ▶ 回顾：给定离散随机变量 X, Y 和函数 g , $Z = g(X, Y)$ 。
- ▶ 定理： $E(Z) = \sum_i \sum_j P(X = x_i, Y = y_j) \cdot g(x_i, y_j)$
- ▶ 给定连续随机变量 X, Y 和函数 g , $Z = g(X, Y)$ 。如何计算 $E(Z)$?
- ▶ 定理： $E(Z) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x, y) \cdot g(x, y) dx dy$
- ▶ 例： $X \sim U(0, 1), Y \sim U(0, 1)$, 且 X 与 Y 相互独立。求 $E(|X - Y|)$
 - ▶ $E(|X - Y|) = \int_0^1 \int_0^1 |x - y| dx dy = \frac{1}{3}$

3. 多维连续随机变量的特征数

► 数学期望的线性性: $E(X + Y) = E(X) + E(Y)$

$$\begin{aligned} & \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x, y) \cdot (x + y) dx dy \\ &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x, y) \cdot x dx dy + \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x, y) \cdot y \cdot dx dy \\ &= \int_{-\infty}^{+\infty} x \cdot \int_{-\infty}^{+\infty} f(x, y) dy dx + \int_{-\infty}^{+\infty} y \cdot \int_{-\infty}^{+\infty} f(x, y) \cdot dx dy \\ &= \int_{-\infty}^{+\infty} x \cdot f_X(x) dx + \int_{-\infty}^{+\infty} y \cdot f_Y(y) dy = E(X) + E(Y) \end{aligned}$$

► 推广: $E(X_1 + X_2 + \cdots + X_n) = E(X_1) + E(X_2) + \cdots + E(X_n)$

3. 多维连续随机变量的特征数

- ▶ 定理：若连续随机变量 X 和 Y 相互独立，则有 $E(XY) = E(X) \cdot E(Y)$
- ▶ $\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x, y) \cdot x \cdot y \, dx dy$
- ▶ $= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f_X(x) \cdot x \cdot f_Y(y) \cdot y \, dx dy$
- ▶ $= \int_{-\infty}^{+\infty} x \cdot f_X(x) dx \cdot \int_{-\infty}^{+\infty} y \cdot f_Y(y) dy$
- ▶ $= E(X) \cdot E(Y)$
- ▶ 推广：若连续随机变量 $X_1, X_2, \dots, X_n$ 相互独立，则有 $E(X_1 X_2 \cdots X_n) = E(X_1) \cdot E(X_2) \cdots E(X_n)$
- ▶ 推论：若连续随机变量 $X_1, X_2, \dots, X_n$ 相互独立，则有 $\text{Var}(X_1 \pm X_2 \pm \cdots \pm X_n) = \text{Var}(X_1) + \text{Var}(X_2) + \cdots + \text{Var}(X_n)$

3. 多维连续随机变量的特征数

- ▶ 例1: 随机向量 $X = (X_1, X_2, \dots, X_n)$ 满足 $X_i \sim N(0,1)$
- ▶ 随机变量 Y 表示 X 的模长。计算 $E(Y^2)$
- ▶ $E(Y^2) = E\left(\sum_{i=1}^n X_i^2\right) = \sum_{i=1}^n E(X_i^2) = n$

3. 多维连续随机变量的特征数

- ▶ 例2: 随机向量 $X = (X_1, X_2, \dots, X_n)$ 满足 $X_i \sim N(0,1)$, 且 X_i 相互独立
- ▶ 给定固定向量 $a = (a_1, a_2, \dots, a_n)$ 。令随机变量 Y 表示 X 与 a 的内积。计算 $E(Y)$ 和 $\text{Var}(Y)$
- ▶ $Y = \sum_{i=1}^n a_i X_i$
- ▶ $E(Y) = E(\sum_{i=1}^n a_i X_i) = \sum_{i=1}^n a_i E(X_i) = 0$
- ▶ $\text{Var}(Y) = \sum_{i=1}^n \text{Var}(a_i X_i) = \sum_{i=1}^n a_i^2$

3. 多维连续随机变量的特征数

- ▶ 例3: $n \times n$ 矩阵 A 每个元素均服从 $N(0,1)$, 且不同元素相互独立
- ▶ 计算 $E(\det(A))$, $E(\text{trace}(A))$, $E(\text{trace}(A^2))$
- ▶ $\det(A) = \sum_{\sigma} (\text{sgn}(\sigma) \cdot \prod_{i=1}^n A_{i,\sigma(i)})$
- ▶ $E(\det(A)) = \sum_{\sigma} \text{sgn}(\sigma) \cdot E(\prod_{i=1}^n A_{i,\sigma(i)}) = \sum_{\sigma} \text{sgn}(\sigma) \cdot \prod_{i=1}^n E(A_{i,\sigma(i)}) = 0$
- ▶ $\text{trace}(A) = \sum_{i=1}^n A_{i,i}$
- ▶ $E(\text{trace}(A)) = E(\sum_{i=1}^n A_{i,i}) = 0$
- ▶ $A_{i,i}^2 = \sum_{j=1}^n A_{i,j} \cdot A_{j,i}$
- ▶ $E(A_{i,i}^2) = \sum_{j=1}^n E(A_{i,j} \cdot A_{j,i}) = 1$
- ▶ $E(\text{trace}(A^2)) = E(\sum_{i=1}^n A_{i,i}^2) = n$

3. 多维连续随机变量的特征数

► 回顾:

- $F(x|y) = \int_{-\infty}^x \frac{f(u,y)}{f_Y(y)} du$ 为给定 $Y = y$ 条件下 X 的**条件分布函数**
- $f(x|y) = \frac{f(x,y)}{f_Y(y)}$ 为给定 $Y = y$ 条件下 X 的**条件密度函数**
- 对于离散随机变量, $E(X|Y = y_j) = \sum_i x_i \cdot P(X = x_i | Y = y_j)$
- 对于二维连续随机变量 X, Y , **定义条件数学期望** $E(X|Y = y) = \int_{-\infty}^{+\infty} f(x|y) \cdot x \cdot dx$
- 回顾: $E(E(X|Y)) = E(X)$

3. 多维连续随机变量的特征数

- ▶ **条件数学期望** $E(X|Y = y) = \int_{-\infty}^{+\infty} f(x|y) \cdot x \cdot dx$
- ▶ $E(E(X|Y)) = E(X)$
- ▶ 例: $X \sim \text{Exp}(\lambda)$, $Y \sim U(0, X)$ 。计算 $E(Y)$ 和 $\text{Var}(Y)$
- ▶ $E(Y|X = x) = \frac{x}{2}$
- ▶ $E(Y) = E(E(Y|X)) = \frac{E(X)}{2} = \frac{1}{2\lambda}$
- ▶ $E(Y^2|X = x) = \frac{x^2}{3}$
- ▶ $E(Y^2) = E(E(Y^2|X)) = \frac{E(X^2)}{3} = \frac{2}{3\lambda^2}$

3. 多维连续随机变量的特征数

- ▶ 给定随机变量 X 和 Y , 定义 X 和 Y 的**协方差**
- ▶
$$\text{Cov}(X, Y) = E\left((X - E(X))(Y - E(Y))\right) = E(XY) - E(X)E(Y)$$
- ▶ 性质回顾:
- ▶
$$\text{Cov}(X, X) = \text{Var}(X) = E(X^2) - (E(X))^2$$
- ▶
$$\text{Cov}(X, Y) = \text{Cov}(Y, X)$$
- ▶
$$\text{Cov}(aX, bY) = ab \cdot \text{Cov}(X, Y)$$
- ▶
$$\text{Cov}(X_1 + X_2, Y) = \text{Cov}(X_1, Y) + \text{Cov}(X_2, Y)$$
- ▶ 若 X 和 Y 相互独立, 则 $\text{Cov}(X, Y) = 0$
- ▶
$$\text{Var}(X_1 + X_2 + \cdots + X_n) = \sum_i \sum_j \text{Cov}(X_i, X_j) = \sum_i \text{Var}(X_i) + 2 \sum_i \sum_{j < i} \text{Cov}(X_i, X_j)$$
- ▶
$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$$

3. 多维连续随机变量的特征数

▶ 例: X, Y 的联合密度函数满足

▶ $f(x, y) = \frac{x+y}{3}$ 若 $0 < x < 1, 0 < y < 2$

▶ 否则 $f(x, y) = 0$

▶ 计算 $\text{Cov}(X, Y), \text{Var}(X), \text{Var}(Y)$

▶ $E(X) = \int_0^1 x \cdot \int_0^2 f(x, y) \cdot dy dx = \int_0^1 \frac{2x+2}{3} \cdot x \cdot dx = \frac{5}{9}$

▶ $E(X^2) = \int_0^1 x^2 \cdot \int_0^2 f(x, y) \cdot dy dx = \int_0^1 \frac{2x+2}{3} \cdot x^2 \cdot dx = \frac{7}{8}$

▶ $E(Y) = \int_0^2 y \cdot \int_0^1 f(x, y) \cdot dx dy = \int_0^2 \frac{2y+1}{6} \cdot y \cdot dx = \frac{11}{9}$

▶ $E(Y^2) = \int_0^2 y^2 \cdot \int_0^1 f(x, y) \cdot dx dy = \int_0^2 \frac{2y+1}{6} \cdot y^2 \cdot dx = \frac{16}{9}$

▶ $E(XY) = \int_0^1 x \cdot \int_0^2 y \cdot f(x, y) \cdot dy dx = \int_0^1 x \cdot \frac{2x+8/3}{3} \cdot dx = \frac{2}{3}$

▶ $\text{Var}(X) = E(X^2) - (E(X))^2 = \frac{13}{162}, \text{Var}(Y) = E(Y^2) - (E(Y))^2 = \frac{23}{81}$

▶ $\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = -\frac{1}{81}$

3. 多维连续随机变量的特征数

- ▶ $X, Y \sim N(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$, 计算 $\text{Cov}(X, Y)$
- ▶ $E(X) = \mu_1, E(Y) = \mu_2$
- ▶
$$f(x, y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp \left[-\frac{1}{2(1-\rho^2)} \cdot \left(\frac{(x-\mu_1)^2}{\sigma_1^2} + \frac{(y-\mu_2)^2}{\sigma_2^2} - \frac{2\rho(x-\mu_1)(y-\mu_2)}{\sigma_1\sigma_2} \right) \right]$$
- ▶ $\text{Cov}(X, Y) = \iint f(x, y) \cdot (x - \mu_1) \cdot (y - \mu_2) dx dy$
- ▶
$$u = \frac{\left(\frac{x-\mu_1}{\sigma_1} - \rho \cdot \frac{y-\mu_2}{\sigma_2} \right)}{\sqrt{1-\rho^2}}, v = \frac{y-\mu_2}{\sigma_2}$$
- ▶ $y = \sigma_2 v + \mu_2, x = \left(\sqrt{1-\rho^2} \cdot u + \rho \cdot v \right) \cdot \sigma_1 + \mu_1$
- ▶ $(x - \mu_1)(y - \mu_2) = \sigma_1\sigma_2 v \left(\sqrt{1-\rho^2} \cdot u + \rho \cdot v \right)$
- ▶
$$f(x, y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp \left[-\frac{1}{2} (u^2 + v^2) \right], \left| \frac{\partial(x,y)}{\partial(u,v)} \right| = \sigma_1\sigma_2\sqrt{1-\rho^2}$$

3. 多维连续随机变量的特征数

- ▶ $(x - \mu_1)(y - \mu_2) = \sigma_1 \sigma_2 v \left(\sqrt{1 - \rho^2} \cdot u + \rho \cdot v \right)$
- ▶ $f(x, y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp \left[-\frac{1}{2} (u^2 + v^2) \right], \left| \frac{\partial(x,y)}{\partial(u,v)} \right| = \sigma_1\sigma_2\sqrt{1-\rho^2}$
- ▶ $\text{Cov}(X, Y) = \iint f(x, y) \cdot (x - \mu_1) \cdot (y - \mu_2) dx dy$
- ▶ $= \iint \frac{1}{2\pi} \exp \left[-\frac{1}{2} (u^2 + v^2) \right] \cdot (x - \mu_1) \cdot (y - \mu_2) du dv$
- ▶ $= \iint \frac{1}{2\pi} \exp \left[-\frac{1}{2} (u^2 + v^2) \right] \cdot \sigma_1 \sigma_2 v \left(\sqrt{1 - \rho^2} \cdot u + \rho \cdot v \right) \cdot du dv$
- ▶ $= \sigma_1 \sigma_2 \iint \frac{1}{2\pi} \exp \left[-\frac{1}{2} (u^2 + v^2) \right] \cdot v \left(\sqrt{1 - \rho^2} \cdot u + \rho \cdot v \right) \cdot du dv$
- ▶ $\iint \frac{1}{2\pi} \exp \left[-\frac{1}{2} (u^2 + v^2) \right] \cdot v u du dv = 0$
- ▶ $\iint \frac{1}{2\pi} \exp \left[-\frac{1}{2} (u^2 + v^2) \right] \cdot v^2 du dv = 1$
- ▶ $\text{Cov}(X, Y) = \rho \sigma_1 \sigma_2$

3. 多维连续随机变量的特征数

- ▶ 给定随机变量 X 和 Y , 若 $\sigma(X), \sigma(Y) > 0$, 定义 X 和 Y 的**相关系数** $\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma(X)\sigma(Y)}$
- ▶ 回顾: $\text{Cov}(X, Y) = E\left((X - E(X))(Y - E(Y))\right)$
- ▶ 回顾: $\tilde{X} = \frac{X - E(X)}{\sigma(X)}$ 为 X 的标准化随机变量
- ▶ $\text{Corr}(\tilde{X}, \tilde{Y}) = E(\tilde{X}\tilde{Y}) = E\left(\frac{X - E(X)}{\sigma(X)} \cdot \frac{Y - E(Y)}{\sigma(Y)}\right) = \frac{\text{Cov}(X, Y)}{\sigma(X)\sigma(Y)} = \text{Corr}(X, Y)$
 - ▶ $\text{Corr}(X, Y) > 0$ (或 $\text{Cov}(X, Y) > 0$) : X 和 Y **正相关**
 - ▶ $\text{Corr}(X, Y) < 0$ (或 $\text{Cov}(X, Y) < 0$) : X 和 Y **负相关**
 - ▶ $\text{Corr}(X, Y) = 0$ (或 $\text{Cov}(X, Y) = 0$) : X 和 Y **不相关**
- ▶ 若相互独立, 一定有不相关
 - ▶ $E\left((X - E(X))(Y - E(Y))\right) = E(XY) - E(X)E(Y) = 0$
- ▶ 不相关, 是否一定有相互独立?

3. 多维连续随机变量的特征数

- ▶ 给定随机变量 X 和 Y , 若 $\sigma(X), \sigma(Y) > 0$, 定义 X 和 Y 的**相关系数** $\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma(X)\sigma(Y)}$
- ▶ 性质1: $|\text{Corr}(X, Y)| \leq 1$
- ▶ 证明:
 - ▶ $g(t) = E \left(\left(t(X - E(X)) + (Y - E(Y)) \right)^2 \right) = t^2 \sigma(X)^2 + 2t \text{Cov}(X, Y) + \sigma(Y)^2$
 - ▶ $g(t) \geq 0, \sigma(X), \sigma(Y) > 0 \Rightarrow (2\text{Cov}(X, Y))^2 - 4\sigma(X)^2 \sigma(Y)^2 \leq 0$

3. 多维连续随机变量的特征数

- ▶ 给定随机变量 X 和 Y , 若 $\sigma(X), \sigma(Y) > 0$, 定义 X 和 Y 的**相关系数** $\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma(X)\sigma(Y)}$
- ▶ 回顾: $\tilde{X} = \frac{X - E(X)}{\sigma(X)}$ 为 X 的标准化随机变量
- ▶ $\text{Corr}(\tilde{X}, \tilde{Y}) = E(\tilde{X}\tilde{Y}) = E\left(\frac{X - E(X)}{\sigma(X)} \cdot \frac{Y - E(Y)}{\sigma(Y)}\right) = \frac{\text{Cov}(X, Y)}{\sigma(X)\sigma(Y)} = \text{Corr}(X, Y)$
- ▶ 性质2: $\text{Corr}(X, Y) = \pm 1 \iff$ 存在 $a \neq 0$ 与 b , $P(Y = aX + b) = 1$
- ▶ 证明 ($\text{Corr}(X, Y) = 1$)
- ▶ $\Rightarrow$: $\text{Var}(\tilde{X} - \tilde{Y}) = \text{Var}(\tilde{X}) + \text{Var}(\tilde{Y}) - 2\text{Corr}(X, Y) = 0$
- ▶ $P(\tilde{X} - \tilde{Y} = c) = 1$
- ▶ $\Leftarrow$: $P(Y = aX + b) = 1, \quad \sigma(Y) = |a|\sigma(X), \quad \text{Cov}(X, Y) = a(\sigma(X))^2$
- ▶ $\text{Corr}(X, Y) = \frac{a}{|a|}$

3. 多维连续随机变量的特征数

► 例1: X, Y 的联合密度函数满足

► $f(x, y) = \frac{x+y}{3}$ 若 $0 < x < 1, 0 < y < 2$

► 否则 $f(x, y) = 0$

► 计算 $\text{Corr}(X, Y)$ 并判断相关性

► $\text{Var}(X) = E(X^2) - (E(X))^2 = \frac{13}{162}, \text{Var}(Y) = E(Y^2) - (E(Y))^2 = \frac{23}{81}$

► $\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = -\frac{1}{81}$

► $\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma(X)\sigma(Y)} = -\frac{\frac{1}{81}}{\sqrt{\frac{13}{162}} \cdot \sqrt{\frac{23}{81}}} = -\sqrt{\frac{2}{299}},$ 负相关

3. 多维连续随机变量的特征数

- ▶ 例2: $X, Y \sim N(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$ 证明 X, Y 相互独立当且仅当 X, Y 不相关
- ▶ X, Y 相互独立等价于 $\rho = 0$
- ▶ X, Y 不相关等价于 $\rho = 0$

4. 多维连续随机变量函数的分布

- ▶ 给定连续随机变量 X, Y 和函数 $g(x, y)$, 求 $Z = g(X, Y)$ 的概率密度函数
- ▶ **卷积公式**: 若 X, Y 相互独立, $Z = X + Y$, 则 $f_Z(z) = \int_{-\infty}^{+\infty} f_X(z - y)f_Y(y)dy$
- ▶ 证明:
 - ▶ $P(Z \leq z) = \iint_{x+y \leq z} f_X(x)f_Y(y)dxdy = \int_{-\infty}^{+\infty} \int_{-\infty}^{z-y} f_X(x)dx \cdot f_Y(y)dy$
 - ▶ $\int_{-\infty}^{z-y} f_X(x)dx = P(X \leq z - y)$
 - ▶ $P(Z \leq z) = \int_{-\infty}^{+\infty} P(X \leq z - y) \cdot f_Y(y)dy$, 两边对 z 求导

4. 多维连续随机变量函数的分布

▶ 例1: $X \sim N(0, \sigma_1^2)$, $Y \sim N(0, \sigma_2^2)$, X, Y 相互独立, 求 $Z = X + Y$ 的概率密度函数

$$\text{▶ } f_Z(z) = \int_{-\infty}^{+\infty} f_X(z-y)f_Y(y)dy = \frac{1}{2\pi\sigma_1\sigma_2} \int_{-\infty}^{+\infty} \exp\left(-\frac{1}{2}\left(\frac{(z-y)^2}{\sigma_1^2} + \frac{y^2}{\sigma_2^2}\right)\right) dy$$

$$\text{▶ } \frac{(z-y)^2}{\sigma_1^2} + \frac{y^2}{\sigma_2^2} = y^2\left(\frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2}\right) - \frac{2yz}{\sigma_1^2} + \frac{z^2}{\sigma_1^2}$$

$$\text{▶ } \text{令 } A = \frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2}$$

$$\text{▶ } \frac{(z-y)^2}{\sigma_1^2} + \frac{y^2}{\sigma_2^2} = A\left(y - \frac{z}{\sigma_1^2 A}\right)^2 + \frac{z^2}{\sigma_1^2} - \frac{z^2}{\sigma_1^4 A} = A\left(y - \frac{z}{\sigma_1^2 A}\right)^2 + \frac{z^2}{\sigma_1^2} \left(1 - \frac{1}{\sigma_1^2 A}\right)$$

$$\text{▶ } \frac{z^2}{\sigma_1^2} \left(1 - \frac{1}{\sigma_1^2 A}\right) = \frac{z^2}{\sigma_1^2 + \sigma_2^2}$$

$$\text{▶ } f_Z(z) = \frac{1}{2\pi\sigma_1\sigma_2} \exp\left(-\frac{1}{2} \cdot \frac{z^2}{\sigma_1^2 + \sigma_2^2}\right) \cdot \int_{-\infty}^{+\infty} \exp\left(-\frac{A}{2}\left(y - \frac{z}{\sigma_1^2 A}\right)^2\right) dy$$

4. 多维连续随机变量函数的分布

- ▶ 例1: $X \sim N(0, \sigma_1), Y \sim N(0, \sigma_2)$, X, Y 相互独立, 求 $Z = X + Y$ 的概率密度函数
- ▶
$$f_Z(z) = \frac{1}{2\pi\sigma_1\sigma_2} \exp\left(-\frac{1}{2} \cdot \frac{z^2}{\sigma_1^2 + \sigma_2^2}\right) \cdot \int_{-\infty}^{+\infty} \exp\left(-\frac{A}{2} \left(y - \frac{z}{\sigma_1^2 A}\right)^2\right) dy$$
- ▶
$$\frac{1}{\sqrt{2\pi} \cdot \sqrt{1/A}} \int_{-\infty}^{+\infty} \exp\left(-\frac{A}{2} \left(y - \frac{z}{\sigma_1^2 A}\right)^2\right) dy = 1$$
- ▶
$$f_Z(z) = \frac{1}{2\pi\sigma_1\sigma_2} \exp\left(-\frac{1}{2} \cdot \frac{z^2}{\sigma_1^2 + \sigma_2^2}\right) \cdot \sqrt{2\pi} \cdot \sqrt{1/A}$$
- ▶ $A = \frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2} \Rightarrow f_Z(z) = \frac{1}{\sqrt{2\pi} \cdot \sqrt{\sigma_1^2 + \sigma_2^2}} \exp\left(-\frac{1}{2} \cdot \frac{z^2}{\sigma_1^2 + \sigma_2^2}\right)$
- ▶ 也即 $Z \sim N(0, \sigma_1^2 + \sigma_2^2)$

4. 多维连续随机变量函数的分布

- ▶ 推广: $X_i \sim N(\mu_i, \sigma_i)$, 且相互独立, 则 $\sum_{i=1}^n a_i X_i \sim N(\mu_0, \sigma_0^2)$
- ▶ $\mu_0 = \sum_{i=1}^n a_i \mu_i$, $\sigma_0^2 = \sum_{i=1}^n a_i^2 \sigma_i^2$
- ▶ 特别有, 若 X_i 独立同分布, 且 $X_i \sim N(0, 1)$, 则 $\sum_{i=1}^n a_i X_i \sim N(0, |a|^2)$
 - ▶ 当 $a_i = \frac{1}{n}$, $\frac{1}{n} \sum_{i=1}^n X_i \sim N\left(0, \frac{1}{n}\right)$

4. 多维连续随机变量函数的分布

▶ 例2: $X \sim \text{Exp}(1), Y \sim \text{Exp}(1)$, X, Y 相互独立, 求 $Z = X + Y$ 的概率密度函数

▶
$$f_Z(z) = \int_{-\infty}^{+\infty} f_X(z-y)f_Y(y)dy = \int_0^z e^{-(z-y)-y} dy = ze^{-z}$$

▶ 也即 $Z \sim \Gamma(2,1)$

▶ 回顾: 对于 $\alpha, \lambda > 0$, 定义概率密度函数

▶
$$f(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}, \text{ 当 } x \geq 0$$

▶
$$f(x) = 0, \text{ 当 } x < 0$$

▶ 推广: $X \sim \Gamma(\alpha_1, \lambda), Y \sim \Gamma(\alpha_2, \lambda)$, X, Y 相互独立, $X + Y \sim \Gamma(\alpha_1 + \alpha_2, \lambda)$

4. 多维连续随机变量函数的分布

- ▶ $X \sim U(0,1), Y \sim U(0,1)$, X, Y 相互独立, 求 $Z = \max\{X, Y\}$ 的概率密度函数
 - ▶ $P(Z \leq z) = P(X \leq z)P(Y \leq z) = z^2$
 - ▶ $f_Z(z) = 2z$ 当 $z \in (0,1)$

- ▶ $X \sim U(0,1), Y \sim U(0,1)$, X, Y 相互独立, 求 $Z = \min\{X, Y\}$ 的概率密度函数
 - ▶ $P(Z \geq z) = P(X \geq z)P(Y \geq z) = (1 - z)^2$
 - ▶ $P(Z \leq z) = 1 - (1 - z)^2$
 - ▶ $f_Z(z) = 2(1 - z)$ 当 $z \in (0,1)$

4. 多维连续随机变量函数的分布

- ▶ 若 $X_1, X_2, \dots, X_n$ 相互独立, 则 $Y = \max\{X_1, X_2, \dots, X_n\}$ 的分布函数 $F(y)$ 满足
 - ▶ $F_Y(y) = F_{X_1}(y) \cdot F_{X_2}(y) \cdot \dots \cdot F_{X_n}(y)$
- ▶ 若 $X_1, X_2, \dots, X_n$ 相互独立, 则 $Y = \min\{X_1, X_2, \dots, X_n\}$ 的分布函数 $F(y)$ 满足
 - ▶ $F_Y(y) = 1 - (1 - F_{X_1}(y)) \cdot (1 - F_{X_2}(y)) \cdot \dots \cdot (1 - F_{X_n}(y))$

4. 多维连续随机变量函数的分布

- ▶ 回顾：设 X 为连续随机变量，若函数 $y = g(x)$ 严格单调，其反函数 $h(y)$ 有连续导数，则 $Y = g(X)$ 的概率密度函数为
 - ▶ $f_Y(y) = f_X(h(y)) \cdot |h'(y)|$ 当 $y \in (\alpha, \beta)$
 - ▶ $f_Y(y) = 0$ 当 $y \notin (\alpha, \beta)$
- ▶ 若连续随机变量 X, Y 的联合密度函数为 $f(x, y)$ 。函数 $u = u(x, y), v = v(x, y)$ 有连续偏导数且 $x = x(u, v), y = y(u, v)$ 为唯一的反函数
- ▶ 则 $U = u(X, Y), V = v(X, Y)$ 的联合密度函数为 $f(x(u, v), y(u, v)) \cdot |J|$ ，其中
 - ▶ $J = \left| \frac{\partial(x, y)}{\partial(u, v)} \right| = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \left(\left| \frac{\partial(u, v)}{\partial(x, y)} \right| \right)^{-1} = \left(\begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} \right)^{-1}$

4. 多维连续随机变量函数的分布

- ▶ 例1: 若 X, Y 相互独立, 且 $X \sim N(\mu, 1), Y \sim N(\mu, 1)$ 。计算 $U = X + Y$ 和 $V = X - Y$ 的联合密度函数
- ▶ $u = x + y, v = x - y \Rightarrow x = \frac{u+v}{2}, y = \frac{u-v}{2}$
- ▶ $J = \left| \frac{\partial(x,y)}{\partial(u,v)} \right| = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{vmatrix} = -\frac{1}{2} \Rightarrow |J| = \frac{1}{2}$
- ▶ $U = X + Y$ 和 $V = X - Y$ 的联合密度函数为
- ▶ $\frac{1}{\sqrt{2\pi}} e^{-\frac{(\frac{u+v}{2}-\mu)^2}{2}} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{(\frac{u-v}{2}-\mu)^2}{2}} \cdot \left| -\frac{1}{2} \right| = \frac{1}{4\pi} e^{-\frac{1}{4}((u-2\mu)^2+v^2)}$
- ▶ 也即 $U \sim N(2\mu, 2), V \sim N(0, 2)$, 且 U, V 相互独立

4. 多维连续随机变量函数的分布

► 例2: 若连续随机变量 X, Y 相互独立, 计算 $U = XY$ 的概率密度函数

► $u = xy, v = y \Rightarrow x = u/v, y = v$

$$\text{► } J = \left| \frac{\partial(x,y)}{\partial(u,v)} \right| = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} 1/v & -\frac{u}{v^2} \\ 0 & 1 \end{vmatrix} = \frac{1}{v} \Rightarrow |J| = \frac{1}{|v|}$$

► U, V 的联合密度函数为 $f\left(\frac{u}{v}, v\right) \cdot |J| = f_X\left(\frac{u}{v}\right) \cdot f_Y(v) \cdot \frac{1}{|v|}$

► U 的边际密度函数为 $\int_{-\infty}^{+\infty} f_X\left(\frac{u}{v}\right) \cdot f_Y(v) \cdot \frac{1}{|v|} \cdot dv$

4. 多维连续随机变量函数的分布

- ▶ 例2: 若连续随机变量 X, Y 相互独立, 计算 $U = XY$ 的概率密度函数
- ▶ U 的概率密度函数为 $\int_{-\infty}^{+\infty} f_X\left(\frac{u}{v}\right) \cdot f_Y(v) \cdot \frac{1}{|v|} \cdot dv$
- ▶ 验证:
- ▶ $P(U \leq u) = \iint_{xy \leq u} f_X(x) f_Y(y) dx dy$
- ▶
$$= \int_0^{+\infty} \int_{-\infty}^{u/y} f_X(x) dx \cdot f_Y(y) dy + \int_{-\infty}^0 \int_{u/y}^{+\infty} f_X(x) dx \cdot f_Y(y) dy$$
- ▶ 对 u 求导, $f_U(u) = \int_0^{+\infty} f_X(u/y) \cdot f_Y(y) \cdot \frac{1}{y} \cdot dy - \int_{-\infty}^0 f_X\left(\frac{u}{y}\right) \cdot f_Y(y) \cdot \frac{1}{y} \cdot dy$
- ▶
$$= \int_{-\infty}^{+\infty} f_X(u/y) \cdot f_Y(y) \cdot \frac{1}{|y|} \cdot dy$$

线性代数入门 (复习)

- ▶ 特征值/特征向量：对于 $x \neq 0$ ，若 $Ax = \lambda x$ ，则 λ 是 A 的特征值， x 是 A 的特征向量
- ▶ 对角化： $A = P\Lambda P^{-1}$ 。 P 的第 i 列为特征值为 Λ_i 的特征向量
- ▶ $\det(A) = \det(\Lambda) = \prod_{i=1}^n \Lambda_i$
- ▶ $\det(A^2) = \det(\Lambda^2) = \prod_{i=1}^n \Lambda_i^2$
- ▶ $\text{trace}(A) = \text{trace}(\Lambda) = \sum_{i=1}^n \Lambda_i$
- ▶ $\text{rank}(A) = \Lambda$ 对角线非零元素数量

- ▶ 对称矩阵： $A_{i,j} = A_{j,i}$
- ▶ 正交矩阵： $UU^T = U^T U = I$ 。 $|Ux| = |x|$
- ▶ 对称矩阵可对角化，且特征值一定为**实数**，不同特征值的特征向量**正交**
- ▶ 对称矩阵的正交分解（谱）： $A = U\Lambda U^{-1} = U\Lambda U^T$

线性代数入门

- ▶ 二次型: $f(x) = x^T A x$, A 为对称矩阵
- ▶ 半正定矩阵: 对于 $x \neq 0$, $x^T A x \geq 0$
- ▶ 正定矩阵, 对于 $x \neq 0$, $x^T A x > 0$

- ▶ 半正定矩阵 $\iff$ 所有特征值均非负
- ▶ 正定矩阵 $\iff$ 所有特征值均为正数

- ▶ 对于半正定矩阵 A , 存在对称矩阵 $B = A^{1/2}$, $B^T B = B^2 = A$
- ▶ 对于正定矩阵, B 可逆, 且 $(B^{-1})^2 = A^{-1}$

- ▶ 对于半正定 (正定) 矩阵 A , 满足 $B^T B = A$ 的 B 不唯一

5. 多维高斯分布

- ▶ 随机向量 $\mathbf{X} = (X_1, X_2, \dots, X_n)$ 。定义 $E(\mathbf{X}) = (E(X_1), E(X_2), \dots, E(X_n))$ 为 $\mathbf{X}$ 的数学期望向量, $\text{Cov}(\mathbf{X}) = E\left((\mathbf{X} - E(\mathbf{X}))(\mathbf{X} - E(\mathbf{X}))^T\right)$ 为 $\mathbf{X}$ 的协方差矩阵

- ▶
$$\text{Cov}(\mathbf{X}) = \begin{pmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) & \dots & \text{Cov}(X_1, X_n) \\ \text{Cov}(X_2, X_1) & \text{Var}(X_2) & \dots & \text{Cov}(X_2, X_n) \\ \dots & \dots & \dots & \dots \\ \text{Cov}(X_n, X_1) & \text{Cov}(X_n, X_2) & \dots & \text{Var}(X_n) \end{pmatrix}$$

- ▶ 性质: $\text{Cov}(\mathbf{X})$ 是半正定的对称矩阵

- ▶ $\alpha^T \text{Cov}(\mathbf{X}) \alpha = E\left(\left(\alpha^T (\mathbf{X} - E(\mathbf{X}))\right)^2\right) = \text{Var}(\alpha^T \mathbf{X}) \geq 0$

5. 多维高斯分布

- ▶ $X_1, X_2 \sim N(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$, 计算协方差矩阵和其逆矩阵

- ▶ 协方差矩阵为 $\mathbf{B} = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}$

- ▶ $\det(\mathbf{B}) = (1 - \rho^2)\sigma_1^2\sigma_2^2$

- ▶ 逆矩阵为 $\mathbf{B}^{-1} = \frac{1}{1-\rho^2} \cdot \begin{pmatrix} \frac{1}{\sigma_1^2} & -\frac{\rho}{\sigma_1\sigma_2} \\ -\frac{\rho}{\sigma_1\sigma_2} & \frac{1}{\sigma_2^2} \end{pmatrix}$

- ▶ $f(\mathbf{x}) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp \left[-\frac{1}{2(1-\rho^2)} \cdot \left(\frac{(x_1-\mu_1)^2}{\sigma_1^2} + \frac{(x_2-\mu_2)^2}{\sigma_2^2} - \frac{2\rho(x_1-\mu_1)(x_2-\mu_2)}{\sigma_1\sigma_2} \right) \right]$

- ▶ $= \frac{1}{2\pi \cdot (\det(\mathbf{B}))^{\frac{1}{2}}} \exp \left(-\frac{1}{2} \cdot (\mathbf{x} - \boldsymbol{\mu})^T \mathbf{B}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right)$

5. 多维高斯分布

- ▶ n 维高斯分布的联合密度函数为

- ▶
$$f(\mathbf{x}) = \frac{1}{(2\pi)^{\frac{n}{2}} \cdot (\det(\mathbf{B}))^{\frac{1}{2}}} \exp\left(-\frac{1}{2} \cdot (\mathbf{x} - \boldsymbol{\mu})^T \mathbf{B}^{-1} (\mathbf{x} - \boldsymbol{\mu})\right)$$

- ▶ 数学期望向量: $\boldsymbol{\mu} \in \mathbb{R}^n$
- ▶ 协方差矩阵: $\mathbf{B} \in \mathbb{R}^{n \times n}$, $\mathbf{B}$ 可逆
- ▶ 记号: $\mathbf{X} \sim N(\boldsymbol{\mu}, \mathbf{B})$

- ▶ 当 $\boldsymbol{\mu} = \mathbf{0}$, $\mathbf{B} = \mathbf{I}_n$

- ▶
$$f(\mathbf{x}) = (2\pi)^{-\frac{n}{2}} \cdot \exp\left(-\frac{1}{2} |\mathbf{x}|^2\right) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}} \cdot \exp\left(-\frac{x_i^2}{2}\right)$$

- ▶ 也即 $X_1, X_2, \dots, X_n$ 相互独立且 $X_i \sim N(0, 1)$

5. 多维高斯分布

- ▶ n 维高斯分布的联合密度函数为

$$\text{▶ } f(\mathbf{x}) = \frac{1}{(2\pi)^{\frac{n}{2}} \cdot (\det(\mathbf{B}))^{\frac{1}{2}}} \exp\left(-\frac{1}{2} \cdot (\mathbf{x} - \boldsymbol{\mu})^T \mathbf{B}^{-1} (\mathbf{x} - \boldsymbol{\mu})\right)$$

- ▶ 性质: $\mathbf{X} \sim N(\boldsymbol{\mu}, \mathbf{B})$ 。令 $\mathbf{Y} = \mathbf{A}\mathbf{X} + \mathbf{b}$, 则有 $\mathbf{A}\mathbf{X} + \mathbf{b} \sim N(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{A}\mathbf{B}\mathbf{A}^T)$

- ▶ 这里要求 $\mathbf{A}$ 行满秩。

- ▶ 证明: 若 $\mathbf{A}$ 是方阵 (可逆)

$$\text{▶ } f_Y(\mathbf{y}) = f_X(\mathbf{A}^{-1}(\mathbf{y} - \mathbf{b})) \cdot \left| \frac{\partial \mathbf{x}}{\partial \mathbf{y}} \right| = f_X(\mathbf{A}^{-1}(\mathbf{y} - \mathbf{b})) \cdot \left| \frac{\partial \mathbf{y}}{\partial \mathbf{x}} \right|^{-1} = f_X(\mathbf{A}^{-1}(\mathbf{y} - \mathbf{b})) \frac{1}{|\det(\mathbf{A})|}$$

$$\text{▶ } = \frac{1}{(2\pi)^{\frac{n}{2}} \cdot (\det(\mathbf{B}))^{\frac{1}{2}}} \exp\left(-\frac{1}{2} \cdot (\mathbf{A}^{-1}(\mathbf{y} - \mathbf{b}) - \boldsymbol{\mu})^T \mathbf{B}^{-1} (\mathbf{A}^{-1}(\mathbf{y} - \mathbf{b}) - \boldsymbol{\mu})\right) \cdot \frac{1}{|\det(\mathbf{A})|}$$

$$\text{▶ } = \frac{1}{(2\pi)^{\frac{n}{2}} \cdot (\det(\mathbf{A}\mathbf{B}\mathbf{A}^T))^{\frac{1}{2}}} \exp\left(-\frac{1}{2} \cdot (\mathbf{y} - \mathbf{A}\boldsymbol{\mu} - \mathbf{b})^T (\mathbf{A}\mathbf{B}\mathbf{A}^T)^{-1} (\mathbf{y} - \mathbf{A}\boldsymbol{\mu} - \mathbf{b})\right)$$

5. 多维高斯分布

- ▶ 例: $X \sim N(\mu, \sigma^2)$, 若 $a \neq 0$, 求 $Y = aX + b$ 服从的分布
- ▶ 例: 若 X_i 独立同分布, 且 $X_i \sim N(0, 1)$, 证明 $\sum_{i=1}^n a_i X_i \sim N(0, |a|^2)$
- ▶ 例: 若 X, Y 相互独立, 且 $X \sim N(\mu, 1), Y \sim N(\mu, 1)$ 。计算 $U = X + Y$ 和 $V = X - Y$ 的联合密度函数

- ▶ 例1: $\mathbf{X} \sim N(\boldsymbol{\mu}, \mathbf{B})$, 给出 $\mathbf{X}$ 的子向量服从的边际分布。给出 X_i 的边际分布
 - ▶ $X_i \sim N(\mu_i, B_{i,i})$

- ▶ 例2: $X, Y \sim N(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$, 求 $X + Y$ 服从的分布
 - ▶ $X + Y \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2 + 2\rho\sigma_1\sigma_2)$

5. 多维高斯分布

- ▶ 一般多维高斯分布可视为独立同分布标准正态分布**线性变换**后的结果
- ▶ $X \sim N(\mathbf{0}, I_n)$, $Y \sim N(\boldsymbol{\mu}, B)$ 。
- ▶ $Y = B^{1/2}X + \boldsymbol{\mu}$
- ▶ $X = B^{-1/2}(Y - \boldsymbol{\mu})$ 。白化 (whitening)

- ▶ 对 $Y \sim N(0, I_n)$ 做线性变换 $f(\mathbf{y}) = A\mathbf{y}$
- ▶ 我们已经通过密度变换公式验证了 $X = AY \sim N(\mathbf{0}, AA^T)$
- ▶
$$f(\mathbf{x}) = \frac{1}{(2\pi)^{\frac{n}{2}} \cdot (\det(AA^T))^{\frac{1}{2}}} \exp\left(-\frac{1}{2} \cdot \mathbf{x}^T (AA^T)^{-1} \mathbf{x}\right)$$

- ▶ 为什么线性变换之后的密度函数只和 $B = AA^T$ 有关?
- ▶ $\det(B)^{1/2}$ 的几何含义? $\mathbf{x}^T B^{-1} \mathbf{x}$ 的几何含义?

5. 多维高斯分布

- ▶ 旋转不变性: $Y \sim N(\mathbf{0}, I_n)$, 若 V 为正交矩阵, 则 $VY \sim N(0, I_n)$
- ▶ 几何理解: Y 的概率密度函数正比于 $\exp\left(-\frac{1}{2}|\mathbf{y}|^2\right)$, 正交变换 V 不改变模长
- ▶ 奇异值分解 (SVD): 对于任意矩阵 A , $A = U\Sigma V$
 - ▶ U 和 V 为正交矩阵
 - ▶ Σ 为对角矩阵, 当 A 可逆时全部对角元素非负
 - ▶ 几何理解: 任何一个线性变换, 都可以分解成旋转 V , 拉伸 Σ , 再旋转 U
- ▶ 由旋转不变性, 只需要关注拉伸 Σ 和第二次旋转 U
- ▶ 拉伸 Σ 和第二次旋转 U 刻画了协方差矩阵 $B = AA^T$ 的谱
- ▶ $B = AA^T = U\Sigma^2 U^T$

5. 多维高斯分布

- ▶ $Y \sim N(\mathbf{0}, I_n)$ 的等密度轮廓线 (equidensity contour) 由欧式空间中的模长刻画
- ▶ $f_Y(\mathbf{y}) \sim \exp\left(-\frac{1}{2}|\mathbf{y}|^2\right)$
- ▶ 线性变换后, $\mathbf{X} = \mathbf{A}\mathbf{y}$ 的等密度轮廓线由协方差矩阵 $\mathbf{B} = \mathbf{A}\mathbf{A}^T$ 刻画
- ▶ $\mathbf{y}^T \mathbf{y} \leq 1 \Rightarrow \mathbf{x}^T (\mathbf{A}\mathbf{A}^T)^{-1} \mathbf{x} \leq 1$
- ▶ 线性变换后的单位球变为了由协方差矩阵 $\mathbf{B}$ 刻画的 **椭球**
- ▶ $\mathbf{B} = \mathbf{A}\mathbf{A}^T = \mathbf{U}\mathbf{\Sigma}^2\mathbf{U}^T$
 - ▶ $\mathbf{\Sigma}$ 刻画了椭球轴长
 - ▶ $\mathbf{U}$ 刻画了椭球轴的方向
 - ▶ $(\det(\mathbf{B}))^{1/2} = \det(\mathbf{\Sigma})$ 刻画了椭球的体积
- ▶ 正则化: 概率密度函数中含有 $(\det(\mathbf{B}))^{1/2}$